Chapter 3
The International System of Units, Traceability, and Calibration



Consistent and well-defined units, much like terminology, are critical in the comparison of measurement results and to the understanding of measurement uncertainty. This chapter provides a history of the International System of Units, or "SI," and a modern description of its seven base units utilized throughout the majority of the world. These units include the second, the meter, the kilogram, the ampere, the kelvin, the mole, and the candela. Recent advancements in unit definitions are discussed, with an emphasis on redefinition using fundamental constants to achieve lower uncertainties. This chapter also describes calibration and traceability, important metrological concepts, along with calibration policies and standards.

3.1 History of the SI and Base Units

The International System of Units, abbreviated "SI" from the French name, Le Système International d'Unités, is the modern form of the metric system. In 1790, Louis XVI of France authorized a scientific investigation to reform the country's weights and measures, which led to the development of the first metric system in 1795. In 1875, 17 countries signed the Metre Convention, which established the Bureau International des Poids et Mesures (BIPM). The BIPM provided a single international system of measurement standards, officially instituting unit definitions for length and mass: the meter and kilogram. The Conference Générale des Poids et Mesures (CGPM), a conference of delegates from around the world, was simultaneously established under the BIPM to periodically evaluate and update the unit definitions and provide guidance on their dissemination to the measurement community. Subsequent meetings of the CGPM added definitions of other units, terminology, the metric prefixes, and measurement guidance. In 1954, the 10th meeting of GCPM declared that a complete measurement system would be derived from 6 base units directly related to mechanical, electromagnetic, temperature, and optical radiation. These six

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base units—meter, kilogram, second, ampere, Kelvin, and candela—would form the basis of the SI. In 1960, the SI was formally enacted by the 11th CGPM.

The SI units continue to evolve based on the decisions of the CGPM. For example, the mole was added as a 7th base unit by the 14th CGPM in 1971 (Bureau International des Poids et Measures (BIPM) 2019). The meter was redefined in 1983 based on the speed of light, rather than the length of a physical artifact. The latest update, approved by the CGPM in 2018, effective in 2019, redefines the kilogram based on the value of Planck's constant rather than the mass of a physical piece of metal known as the International Prototype of the Kilogram (IPK). This latest update changed the definitions of the kilogram, ampere, mole, and Kelvin; all SI units are now defined based on physical constants such as the speed of light and electron charge, rather than on artifacts that are prone to damage or drift over time. This redefinition also established some additional physical constants, such as the electron charge, as fixed values with no uncertainty. In other words, the value of these constants is determined based on the best consensus value between metrologists. Prior to the redefinition, physical constants were empirical values, with uncertainty, based on measurements using the unit definitions at the time.

The base units can be thought of as the fundamental set of seven units from which all other SI units can be derived. The SI and base units are part of the same basic measurement language used internationally. The seven base units are defined below (by their 2019 definitions) along with the seven fundamental physical constants that define the units. A list of some common derived units is also provided. The definitions in italics are quoted from the 9th edition of the SI Brochure (Bureau International des Poids et Measures (BIPM) 2019).

3.1.1 SI Constants

In defining the seven base units, the SI relies on several physical constants that are fixed with no uncertainty, as shown in Table 3.1. These constants are used to define the SI units.

Table 3.1 Physical constants defining the SI (Bureau International des Poids et Measures (BIPM) 2019)

Defining Constant	Symbol	Value	Unit
Hyperfine transition frequency of Cs-133	$\Delta\nu_{Cs}$	9,192,631,770	s^{-1}
Speed of light in vacuum	c	299,792,458	$m\ s^{-1}$
Planck constant	h	$6.62607015 \times 10^{-34}$	$kg\ m^2\ s^{-1}$
Elementary charge	e	$1.602176634 \times 10^{-19}$	A s
Boltzmann constant	k	1.380649×10^{-23}	$kg\ m^2\ s^{-2}\ K^{-1}$
Avogadro constant	N_A	$6.02214076 \times 10^{23}$	mol^{-1}
Luminous efficacy	K_{cd}	683	$cd\ kg^{-1}\ m^{-2}\ s^3$

3.1.2 Time: Second (s)

The second is the duration of 9,192,631,770 ($\Delta\nu_{Cs}$) periods of radiation corresponding to the transition between two hyperfine levels of the cesium-133 atom ground state.

Many definitions of the second have been used or proposed throughout metrological history, ranging from measuring the period of a one-meter pendulum to dividing the time between full moons. Currently based on atomic vibrations, atomic clocks are now in use with an uncertainty of less than 1 s in 100 million years (Ball 2013).

3.1.3 Length: Meter (m)

The meter is the length of the path travelled by light in vacuum during a time interval of 1/299,792,458 (1/c) of a second.

The first meter was defined in 1795 to be one ten-millionth the circumference of the Earth. A metal bar was created a few years later based on this length and the meter was redefined to be the length of the bar. Some form of bar was used as the standard until the 1960s. In 1983, the current definition based on the speed of light was adopted. Devices called interferometers can convert the speed of light into an actual meter of length.

3.1.4 Mass: Kilogram (kg)

The kilogram is the mass found by taking the value of the Planck constant, h, of $6.62607015 \times 10^{-34}$ kg m^2 s^{-1}, given the definition of the meter and second [given above].

The first definition of a kilogram was the mass of one liter of pure water. Because it was difficult to replicate this definition under different conditions, in 1799 the kilogram was defined based on a platinum artifact. This artifact was formally replaced in 1889 with a platinum-iridium cylinder, known as the IPK and maintained by the BIPM in a vault near Paris. When the IPK was made, many duplicates were also manufactured and sent to different National Metrology Institutes (NMIs). However, measurements on the IPK and its siblings now show microgram-level discrepancies in their mass values, prompting the BIPM's 2019 redefinition of the kilogram to its current form based on Planck's constant. A Kibble balance (see Sect. 3.4.1) can convert this definition into a kilogram of mass.

For historical reasons, the kilogram is the only base unit that includes a metric prefix: "kilo." The original base unit was the gram, but the small mass of one gram was difficult to fabricate and work with, so the defining artifact was made one kilogram. Eventually, the definition was updated to consider a kilogram the base unit, so the kilogram artifact could be used directly.

3.1.5 Electric Current: Ampere (A)

The ampere is that constant current corresponding to the flow of 1/1.602176634 × 10^{-19} elementary charges (1/e) per second.

Prior to 2019, the elementary charge was an empirical value with some uncertainty. The 2019 SI definition fixes the elementary charge as an exact quantity and bases the Ampere on that value.

3.1.6 Temperature: Kelvin (K)

The kelvin is equal to the change of thermodynamic temperature that results in a change of thermal energy kT by 1.380649 × 10^{-23} J (k).

Previous definitions of the Kelvin were based on the triple-point of water: the temperature and pressure at which water can exist as a solid, liquid, and gas simultaneously. Obtaining pure water to realize this definition was difficult, so the 2019 change redefines the Kelvin based on a value of the Boltzmann constant, *k*.

3.1.7 Quantity of Substance: Mole (mol)

The mole is the amount of substance of a system that contains 6.02214076 × 10^{23} (N_A) elementary entities.

The mole is a measure of the amount of a substance and is mostly used in chemistry. Unlike the other units, it is simply a number (Avogadro's constant) based on the number of atoms in 12 g of carbon-12. The mole has been an SI base unit since 1971, although there was and remains controversy in the metrology community over whether the mole is truly a base unit or simply a scaling factor (Josephson 1962).

3.1.8 Luminous Intensity: Candela (cd)

The candela is the luminous intensity, in a given direction, of a source that emits monochromatic radiation of frequency 540 × 10^{12} Hz and that has a radiant intensity in that direction of 1/683 watt per steradian (1/K_{cd}).

The original definition of the candela relied on the intensity of black body radiation emitted from platinum at its melting point. The current definition was adopted in 1979 to avoid reliance on a specific material. One candela is approximately the brightness of a typical candle flame.

Table 3.2 Examples of derived units (Bureau International des Poids et Measures (BIPM) 2019)

Derived quantity	Special SI name	Base units
Area		m^2
Volume		m^3
Speed, velocity		m/s^{-1}
Acceleration		m/s^{-2}
Density		$kg\ m^{-3}$
Frequency	Hertz (Hz)	s^{-1}
Force	Newton (N)	$kg\ m\ s^{-2}$
Pressure	Pascal (Pa)	$kg\ m^{-1}\ s^{-2}$
Energy	Joule (J)	$kg\ m^2\ s^{-2}$
Voltage	Volt (V)	$kg\ m^2\ s^{-3}\ A^{-1}$
Dose equivalent	Sievert (Sv)	$m^2\ s^{-2}$

3.2 Derived Units

All other measurement units can be formed by combining the base units stated in Sect. 3.1. Derived units are products of powers of the base units. For example, the SI unit for area is meters squared, m^2. The SI also defines the prefixes kilo-, mega-, milli-, micro-, etc., so that km^2 is also an SI unit. Some derived units are given special names by the SI; for example, the newton is an SI unit of force defined as $kg\ m\ s^{-2}$ in base units. Examples of common derived units, including special names given by the SI, if applicable, appear in Table 3.2.

While discouraged by the BIPM, use of imperial and United States customary units (feet, inches, pounds) is still common in USA manufacturing and industry. In 1959, the few countries still using this unit system signed an agreement to define the yard and pound based on an exact number of meters and kilograms, respectively (Mueller and Astin 1959), effectively making the imperial system an SI-derived system.

3.3 Unit Realizations

The units described in Sect. 3.1 are given abstract definitions based on fundamental constants. To be useful for measurement, the units must be "realized" by converting the abstract definition into a physical value that can be used for comparison. The BIPM offers guidance on experimental setups that can realize the units. Based on the current definition of units, anyone can, in theory, construct one of these experiments to realize a unit without reliance on a higher organization such as BIPM or the National Institute of Standards and Technology (NIST) to provide traceability. As technology progresses, new experimental methods may be developed that realize the units with lower uncertainty or reduced cost.

This section provides two examples of unit realization for length and voltage.

3.3.1 Gauge Block Interferometer

The meter is based on the distance light travels in a specific amount of time. To be useful for calibration, this definition must be transferred to a set of master gauge blocks. Gauge blocks are standards used to obtain SI traceability in dimensional metrology. A Gauge Block Interferometer (GBI) is a device used to realize the meter. GBIs are instruments used worldwide for the highest accuracy calibration of gauge blocks and similar metrology standards. Two frequency-stabilized lasers at different wavelengths are reflected off the gauge block and a reference mirror. The interference of the two lasers creates a fringe pattern that is used to determine phase difference and thus length difference of the two beam paths. This difference can determine the length of the gauge block and thus realize the meter.

3.3.2 Josephson Volt

The SI unit for electric potential, the volt, can be realized using the Josephson effect. In 1962, Brian Josephson predicted quantum effects that occur when electron pairs tunnel between two superconductors separated by a thin insulating barrier now known as a Josephson Junction (Josephson 1962). The I–V curve predicted by Josephson includes regions of constant voltage V_n occurring at values $nhf/(2e)$, where n is an integer representing the voltage step, h is Planck's constant, and e is the elementary charge of the electron. e and h are both fundamental constants defined by the SI (see Table 3.1). The ratio $2e/h$ is referred to as the Josephson constant, K_J. Prior to 2019, the Josephson constant was an empirical value with some uncertainty. Under the 2019 SI redefinition, K_J is an exact value with no uncertainty. To differentiate between the two values, K_J is used to represent the value under the 2019 SI redefinition, while $K_{J\text{-}90}$ is used to represent the conventional value defined by CGPM in 1990.

A Josephson voltage standard uses a superconductive integrated circuit chip containing thousands of Josephson junctions in a series–parallel configuration. When cooled to 4 K and biased with microwave frequencies, the chip becomes superconducting and will generate stable voltages that depend only on the applied frequency and K_J constant. The Josephson voltage standard is the most accurate method to generate or measure voltage and is the universal basis for voltage measurements.

3.4 Advancements in Unit Definitions

Ongoing research in the field of metrology is continually improving measurement techniques which, in turn, affect the BIPM's unit definitions. This section gives two examples of advancement in measurement science. First, the Kibble Balance was key in redefining the kilogram to be based on Planck's constant. Second, the intrinsic pressure standard is a new method of realizing the pascal over a large pressure range with much lower uncertainty.

3.4.1 Kibble (Watt) Balance

In 1999, the CGPM formally declared that the kilogram should be redefined without need for a physical artifact (the IPK). Several competing methods were considered for this redefinition, but eventually the Kibble Balance proved the method of choice. The Kibble Balance provides a method of realizing the value of the kilogram in terms of current and voltage. Since current and voltage can be defined in terms of fundamental physical constants, such as the speed of light and Planck's constant, they provided a way to define the kilogram in terms of constants rather than the IPK, which is vulnerable to deterioration or damage and can drift. In 2018, the CGPM approved the new definition of the kilogram, with use of the Kibble Balance to realize the unit (Bureau International des Poids et Measures (BIPM) 2019).

The Kibble Balance was originally called a "Watt" balance since it performs measurements of both current and voltage in the coil, the product of which is expressed in watts, the SI unit of power. This product equals the mechanical power of the test mass in motion. The balance was renamed in honor of its inventor, Bryan Kibble, after he passed away in 2016.

The Kibble Balance is an extremely accurate device for measuring mass. Like conventional balances, it is designed to equalize one force with another, but on a Kibble Balance, the weight of the test mass is offset by the force produced by an electromagnetic field. An upward force is exerted on the electromagnet coil when its field interacts with the surrounding magnetic field. By adjusting the applied current, the magnitude of the force can be controlled and adjusted as necessary to support the weight of the test mass. While similar electromagnetic balances have been used for many years, the Kibble Balance adds a second measurement: using the balance at constant velocity. By swinging the balance arms at a constant velocity, the electrical power can be related to mechanical power. When combined with the static measurement of electrical power required to equalize the balance, the mass of the object can be accurately determined (Kilogram: The Kibble Balance 2018).

The first step to measure a mass with a Kibble Balance is to use the "weighing" or "force" mode. In this mode, a test mass is placed on the mass pan. It exerts a downward force equal to its mass (m) times the local gravitational field(g); $F = \text{mg}$. Current applied to the electromagnetic coil creates an upward force that balances the downward force of the mass. The force (F) on the coil is equal to the current (I) times the magnetic field (B) times the length of the wire in the coil (L); in other words, $F = \text{IBL}$. Combining these two forces, one can see that $mg = \text{IBL}$.

The second measurement, using the "velocity" mode, is necessary because BL is extremely difficult to measure. Adding this measurement allows the measurement of BL to be circumvented. In this mode, the test mass is removed from the mass stage and the current removed from the coil. The coil is then moved through the surrounding field at a controlled velocity. The resulting voltage (V) is measured. Here, $V = \text{vBL}$ where v is the velocity, and B and L are the same field strength and wire length as in weighing mode.

Combining the equations from weighing mode and velocity mode, the mass can be calculated as $m = IV/gv$. Here, current and voltage can be determined using quantum-electrical effects that are measurable on laboratory instruments. Both current and voltage are defined in terms of h, Planck's constant, and e, the charge of the electron.

3.4.2 Intrinsic Pressure Standard

The current method of realizing the pascal, the derived SI unit for pressure, uses mercury manometers. To improve uncertainties, reduce measurement time, and eliminate the need for hazardous mercury, an intrinsic pressure standard is being developed based on highly accurate optical interferometry to link the pascal to the refractive index of helium (Ahmed et al. 2016). The refractive index of a gas depends on its density, which is a function of temperature and pressure. Using the relationship between refractive index, temperature, and pressure for atomic helium, it is possible to determine the refractive index to an accuracy better than one part in 10^6. This will significantly decrease measurement uncertainties and could drastically improve uncertainties in pressure measurement. The same method could improve the realization of Kelvin.

To implement this pressure realization, it is necessary to measure the refractive index with extremely high accuracy, which is possible using laser interferometry. A Fixed-Length Optical Cavity is being developed by NIST to measure the difference in light passing through two channels: one under vacuum and the other filled with gas at the pressure to be measured. The presence of gas in the one cavity will slow the light by an amount relative to the pressure and refractive index. After the light passes through the two cavities it is combined to generate an interference pattern that can be used to calculate the difference in optical length and then calculate the pressure of the gas (Lee 2019).

This new mercury-free intrinsic pressure standard promises to improve pressure and temperature uncertainties. The optical pressure standard is expected to become a commercial device that is comparably priced to commercial pressure standards. The optical pressure standard should provide a large pressure range from a planned 1 Pa to 360 kPa along with lower uncertainties (Hendricks et al. 2014).

3.5 Metrological Traceability

Metrological traceability is obtained through an unbroken chain of calibrations, with uncertainty evaluated at each step, linking them to a relevant primary standard of the SI unit of measurement. The link to SI units may be achieved by reference to national measurement standards. National measurement standards may be primary standards, which are primary realizations of the SI units, or representations of SI units based on

fundamental physical constants, or secondary standards that are calibrated by another NMI. Traceability to the SI between the standards used to perform calibrations and the unit being calibrated must be documented. Note that metrological traceability establishes the units of measurement that express a measurement result. Metrological traceability requires an established calibration hierarchy, and each step in the hierarchy chain has its own contribution to the measurement uncertainty.

When there is a measurement model with more than one input quantity, each of these input quantities should be metrologically traceable and the calibration hierarchy may form a network rather than a chain (JCGM 200 2012). When a calibration is performed using imperial units, traceability can be established using the fixed conversion factors between the imperial units and the SI units.

3.6 Measurement Standards

Measurement standards are artifacts used to define and realize measurement units and are the basis for all lower-echelon measurements. Standards are the fundamental reference for any system of weights and measures against which all other measuring devices are compared. There is a three-level hierarchy of physical measurement standards that are typically used to realize an SI unit.

1. "Primary standards" are at the top level. Primary standards have the highest metrological quality and are the realization for the unit of measurement.
2. "Secondary standards" are at the second level. Secondary standards are calibrated with reference to a primary standard through an unbroken chain to achieve metrological traceability to the SI.
3. "Working standards" are the lowest tier of a standard. Working standards are typically used to calibrate commercial and industrial measurement equipment (de Silva 2002).

An example of a measurement standard used in a calibration laboratory is a Thomas-type 1-ohm resistor commonly used as a primary standard for calibrating secondary resistances.

3.6.1 Certified Reference Materials

Certified Reference Material (CRM) is a type of measurement standard made of discrete quantities of a known substance or minor artifact that has been certified based on its composition, purity, concentration, or some other characteristic. CRMs are essential in stoichiometry, the metrology of chemistry, and are also finding much use in the growing field of nanotechnology. Commercially available CRMs include ores, pure metals and alloys, gases and gas mixtures, nanoscale particles and structures, and many biochemical substances and organic compounds.

Purchased CRMs have their properties certified by a procedure that establishes traceability to an accurate realization of the appropriate unit. The certificate should also include an uncertainty at a stated level of confidence. Certified reference materials that are provided by a supplier are considered traceable when the supplier complies with ISO Guide 34, as accredited by an organization that currently holds mutual recognition signatory status under the International Laboratory Accreditation Cooperation (ILAC) Mutual Recognition Arrangement (MRA) (Sharp 1999). The term "Standard Reference Material" is a trademark applied to CRMs sold by NIST that meet additional documentation requirements (May et al. 2000).

An example of a CRM is a film thickness standard consisting of a quartz substrate with precise and traceable 180-nm deep grooves etched into the surface. The step height of the grooves has been calibrated by the manufacturer and is traceable through NIST. The standard is used to calibrate the thickness of a thin film on a substrate used as a secondary standard to calibrate equipment used for measuring film thickness with X-ray fluorescence.

3.6.2 Check Standards

A check standard is a reference artifact that can be used to closely monitor the status of a system or piece of M&TE. A check standard should be periodically measured using the system and the results should be recorded in a control chart to evaluate the health of a system. This is not the same as a calibration; a check standard needs to be stable over time with little drift, but not necessarily calibrated. The check standard ensures that a measurement is being carried out correctly and that the system is working as intended.

Check standards are a great way to expose errors that afflict a process over time. Once a baseline for the quantities has been established based on the check standard's historical data, they can be used to control the bias and long-term variability of the process.

Check standards must be similar to the test items measured on the system and ideally should be actual test items from the system. Check standards should always be stable and available to the measurement process. If the check standard is measured regularly and at regular intervals, it can help ascertain the true output of the measurement process by determining the part-to-part variation of each measurement sequence (NIST/SEMATECH 2013).

An example of a check standard is a set of microwave attenuators, 3 dB, 40 dB, and 80 dB, maintained by the Microwave Lab at the Sandia Primary Standards Laboratory. These attenuators are measured before every calibration to ensure the measurement equipment is configured correctly and that cables and connectors are functioning as expected.

3.7 Calibration

Calibration is the process of comparing an unknown value of interest to a known traceable value. Calibration is critical because the accuracy of all measuring devices degrades over time. Calibration ensures that equipment is operating in accordance with its certified specifications throughout its operating life. Ensuring that the equipment can measure accurately and agree with measurements around the world ensures product quality.

In a calibration, a measurement standard provides a known traceable value and the device under test (DUT) provides the unknown value. Calibration allows users to determine the uncertainty of a DUT and ensures that the measurements performed using that DUT are metrologically traceable. The accepted method for determining uncertainty in a calibration is defined by the GUM (JCGM 200 2012) and is described in detail in Chaps. 6, 7, and 8.

A proper calibration results in a report of the DUT's value, uncertainty, and the confidence level in that uncertainty. For example, a calibration report for a resistor may state that the resistor value is 1.000 Ω ± 35 μΩ/Ω at 95% confidence. The report should list any measurement standards used for the calibration and provide a statement establishing metrological traceability to the SI. This is typically ensured by periodically calibrating the measurement standards using an accredited calibration provider or NMI, first assuring the provider is accredited for the desired calibration.

The calibration report from commercial calibration laboratories typically provides a "time-of-test uncertainty," including only uncertainty in the measurement made on the DUT the day it was calibrated. Some labs, including the Primary Standards Laboratory at Sandia, provide a calibration certificate with a "certification uncertainty." This value includes additional uncertainty for aspects such as drift, shipping, and handling to account for use of the device during the upcoming interval.

Tolerance testing can be a form of calibration. The result of a tolerance test is a pass or fail determination based on whether the device value falls within an acceptable range. No uncertainty statement is provided in the report. To perform a tolerance test, the measurement standard must have uncertainty at least four times better than the DUT's required tolerance. The mathematical basis for this requirement comes from a risk evaluation and is described in Chap. 5.

3.7.1 The Calibration Cycle

Calibration must be completed periodically to ensure a DUT is operating as expected throughout its lifetime. The recalibration interval, often one year, is determined by the calibration lab with input from the customer and ensures that drift or degradation in the DUT is accounted for before it becomes problematic in use. If results of a calibration indicate the DUT value has changed by more than the previous

calibration's expanded uncertainty, or if a tolerance test results in a fail condition, the DUT is considered out of tolerance (OOT). An OOT indicates that at some point during the last interval, the DUT drifted outside of its certification uncertainty. When this occurs, all measurements made using that DUT during the past interval are questionable and should be analyzed for the possible impact of a bad measurement.

Some DUTs can be adjusted at the time of calibration, but adjustment is not the same as calibration. Before an adjustment is made, a calibration must be performed to determine the "as-found" condition of the DUT. Then, the adjustment is made to bring the DUT closer to the desired nominal value. Finally, a second calibration is performed to determine the "as-left" condition, which can be used until the next calibration date.

At the end of a DUT's life cycle, when it is no longer needed for metrologically traceable measurements, a close-out calibration is performed. This ensures that the equipment functioned as expected during the final calibration interval. If the as-found close-out calibration is OOT, an impact analysis should be performed to determine if the failure impacted the equipment's measurements during the calibration interval.

3.7.2 Legal Aspects of Calibration

"Legal metrology" refers to legal requirements related to metrology used to control the daily transactions of trade and commerce around the world. Legal metrology exists to ensure fair trade and the safety and health of the public and the environment. Domestic uniformity in legal metrology is the responsibility of NIST through its Office of Weights and Measures. Internationally, it is handled by an international agreement and a quasi-official body, the International Organization of Legal Metrology (OIML). Examples of legal metrology include ensuring that fuel purchased by volume is measured using calibrated pumps through all parts of the distribution chain, emissions from power plants are monitored using calibrated equipment to ensure compliance with environmental laws, and municipal water sources use calibrated measurements to evaluate contaminant levels in the water supply. Recall the "Deflategate" case study from Chap. 1 and consider how calibration (or lack thereof) could affect the outcome of civil and criminal investigations when measurements are used as evidence.

3.7.3 Technical Aspects of Calibration

M&TE must be calibrated with consideration of customer requirements and needs. If a calibration is not performed in a similar fashion to how the equipment is used, the results of the calibration are not valid, and the certification does not apply to the measurements being performed. An example is the difference between pulsed

measurements and AC or the difference between DC and AC. If the customer is using the equipment at AC frequencies, the equipment shall be calibrated at the same AC frequencies for the range of use.

Always consider measurement uncertainty, both when performing a calibration and when using the M&TE. When using the DUT to determine compliance to a specification, the DUT uncertainty must always be at least 4 times better than the required tolerance of the measurement. In some cases, a DUT uncertainty as low as 1.5 times better than the requirement may be acceptable if proper guardbanding techniques are used to mitigate the risk of a lower-quality measurement (see Chap. 5.) The quality of the measurements may affect the related risk associated with decisions made regarding the measurement.

3.7.4 Calibration Policies and Requirements

The calibration certificate provided by an outside lab must be trustworthy and the measurements certified in the calibration must be metrologically traceable. Accreditation gives equipment owners greater confidence that the calibration has been performed correctly. Accreditation helps ensure that the calibration process has been reviewed and complies with internationally accepted technical and quality metrology requirements. Calibration laboratories are accredited to the international metrology standard ISO/IEC 17025:2017 (ISO/IEC 17025 2017). In addition, the American National Standard ANSI/NCSL Z540.3:2006 (ANSI/NCSLI 2006) is the American standard listing general requirements for the competency of calibration laboratories.

All accredited calibration laboratories must be periodically assessed by an independent accreditation body such as the National Voluntary Laboratory Accreditation Program (NVLAP) or American Association for Laboratory Accreditation (A2LA). To obtain accreditation status, a calibration lab must demonstrate the ability to perform acceptable calibrations, have a defined scope that documents their measurement capabilities, and have documented uncertainty analyses of their approved measurement systems to ensure they can meet their defined scope. Accreditation assessments require that the laboratory can demonstrate they can make traceable measurements.

3.7.4.1 ISO 17025

ISO/IEC 17025: *General Requirements for the Competence of Testing and Calibration Laboratories* is the recognized worldwide ISO standard for testing and calibration laboratories. For a laboratory to be deemed technically competent in calibration, it must either hold accreditation to ISO/IEC 17025 or at least comply with the requirements in the document. ISO/IEC 17025 gives calibration laboratories one standard to demonstrate that they can operate competently and provide valid

results that are metrologically traceable. Having one international standard enables a wider acceptance of results between countries, ensuring that when a measurement is performed in one part of the world, it can be repeated accurately in another.

A revision to ISO/IEC 17025 was released in 2017. Calibration labs have until November 2020 to fully implement the new standard, which emphasizes taking a risk-based approach to calibration, to maintain their accreditation status.

3.7.4.2 ANSI Z540.1 and ANSI/NCSL Z540.3:2006

ANSI/NCSL Z540.1 and ANSI/NCSL Z540.3:2006 are calibration standards facilitated by the ANSI and the National Conference of Standards Laboratories International (NCSLI). ANSI Z540.1 was originally used to accredit calibration laboratories and provides guidance for general competency of a calibration lab, including requirements on maintaining records, quality assurance, and personnel. ANSI/NCSL Z540.3:2006 provides the requirements for the calibration of an organization's equipment when used in the manufacturing, modification, or testing of products. This promotes confidence in calibration laboratories and ensures that M&TE can be operated in compliance with its certified specifications. One of the ANSI/NCSL Z540.3:2006's major requirements is that decisions based on measurements from calibrated equipment must result in a false acceptance rate of less than 2%. Chapter 5 provides more information on determining false acceptance rates.

3.8 Summary

The International System of Units, or SI, is central to metrology and essential for making credible estimates of measurement uncertainty. The SI constitutes what we define as *real* in the world of measurement. Traceability of measurements to the SI is necessary for making meaningful estimates of measurement uncertainty. Calibration by an accredited laboratory is a means of providing this traceability. While any vendor, laboratory, or individual can claim traceability, without accreditation from a reputable body, it is up to the end-user of M&TE to prove it.

Calibration has associated technical, quality, and legal aspects making it an industry in and of itself. Expertise in many aspects of calibration takes years of hands-on experience. Calibration technicians constitute a specialized field in both military services and the commercial sector.

The SI is not static, as seen by its long history. Modernizations have been necessary to reduce measurement uncertainties for advanced manufacturing and technology requirements. While many in the USA may claim they do not use the SI, for metrology purposes modern U.S. customary units are effectively derived from the SI.

3.9 Related Reading

As seen in this chapter, the SI is a constantly evolving system. With redefinition of four (out of seven) base units in 2019 alone, it is clear that improvements to the system of base units are at the forefront of research conducted by National Metrology Institutes in the quest for reducing uncertainties in measurement. For this reason, we recommended that the reader check the BIPM website (https://www.bipm.org) for the latest version of the SI brochure. NIST also offers additional documentation on the fundamental system of units (Newell and Tiesinga 2019).

The research and development efforts that went into the replacement of the artifact defining the kilogram (IPK) were tremendous and cannot be adequately described here. For further reading on the development and theory behind the Kibble balance, Robinson and Schlamminger (2016) offer a comprehensive review of the new SI definition for the unit of mass and include historical perspectives.

Other modernizations of SI units are equally as interesting, and in some cases, controversial. In modern times, temperature has been realized using the International Temperature Scale. Preston-Thomas (1990) provides an English translation of the currently used ITS-90. While ITS-90 is still used to realize temperature, the redefinition of the Kelvin in 2019 is considered by some to be the most radical change in the SI to date. Machin (2018) provides a backdrop to the unit's redefinition, concisely describing more detailed technical aspects motivating the redefinition.

The full history of the International System of Units cannot be adequately covered in this chapter. Previously introduced in Chap. 1, Klein (1988) provides a well-written, in-depth history with the aforementioned gaps in recent developments.

Interlaboratory comparisons (ILCs) and proficiency tests (PTs) are an important part of metrology and crucial to ensure the validity and quality of accredited calibration laboratories. For more detail on how ILCs and PTs are conducted, ISO/IEC 17043:2010 (ISO/IEC 17043 2010) provides an overview of requirements. NIST also provides more specific general guidelines in the NVLAP accreditation program (Merkel and White 2016) along with discipline-specific guidance in conducting PTs and ILCs (Gust and Harris 2005).

While this book serves as the perfect guide for the metrologist in applying statistics and conducting uncertainty analyses, it does not serve as a reference for training in calibration fundamentals for different disciplines. *The Metrology Handbook* (Bucher 2012), mentioned in Chap. 1, is a good reference for the calibration technician or calibration laboratory manager, in covering detailed quality requirements and principles. *Calibration: A Technician's Guide* (Cable 2005) offers foundational instruction for either the benchtop or field technician performing calibrations on M&TE. For discipline-specific calibration references, the second edition of *Calibration*: *Philosophy and Practice* (Spang 1994) from Fluke provides comprehensive information regarding calibration of electrical instrumentation, especially instrumentation operating in the DC and low frequency regimes. For dimensional calibration, Mitutoyo's *Metrology Handbook: The Science of Measurement* delivers coverage of numerous dimensional metrology topics for those performing calibration.

3.10 Exercises

1. How did the ancient Egyptians define the unit of length? List three potential issues with using and applying this unit in practice.
2. List three units still in modern usage derived from measures of the human body. Why do you think these are still used?
3. The *cord* is a unit of measure of volume used for firewood sales in many countries around the world.
   a. How is this unit standardized in the USA?
   b. What is its equivalent to in cubic meters?
   c. What are some inherent sources of variability in this unit?
4. The time constant for an electrical circuit is an important parameter in determining the dynamic response of the system. For the following cases, show how the units on the left-hand side and right-hand side of the equation are equivalent (hint, derive in terms of base units).
   a. In a simple RC circuit, the time constant is $\tau = RC$. Here τ is in seconds, R is in ohms, and C is in farads.
   b. In a simple RL circuit, the time constant is $\tau = \frac{L}{R}$. Here τ is in seconds, R is in ohms, and L is in henrys.
5. The British thermal unit (Btu) is a non-SI unit of energy in common usage in the USA. How is the Btu defined? Has it always been defined this way? What issues do you see with having a unit whose definition has changed over time?
6. The typical heat flux in a nuclear reactor is 1595 Btu/(minute $\times$ ft^2). Express this in terms of SI units.
7. The SI unit for force is the newton. Show how this can be expressed in terms of base units.
8. The electron volt (eV) is the amount of kinetic energy gained by an electron accelerating from rest through a constant potential difference of 1 V in vacuum. Based on the 2019 redefinition of base units, derive the value of the electron volt in terms of it's SI equivalent, the joule (J).
9. The sale of many produce items at grocery stores and supermarkets is by mass. In most localities, these scales are required to be calibrated. Draw a traceability chain, going back to the fundamental definition of the kilogram, for the measured mass of bananas purchased at a grocery store.
10. The previous definition of the kilogram was set as the mass of the International Prototype of the Kilogram (IPK). 40 prototype kilograms in total were manufactured in 1882 from a platinum-iridium alloy (90% platinum, 10% iridium) and distributed to National Metrology Institutes around the world. Calibration of these prototypes against the IPK showed an average relative

mass drift of 50 µg from 1889 to 1989, with some units drifting by more than 100 µg over this period.

a. Why is this drift concerning? Discuss the implications for daily measurements of mass in commerce or in a laboratory environment.
b. Research the cause of this drift and propose three possible explanations. What could be done to fix the issue(s)?
c. Discuss advantages and disadvantages of redefining the kilogram in terms of fundamental constants.

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